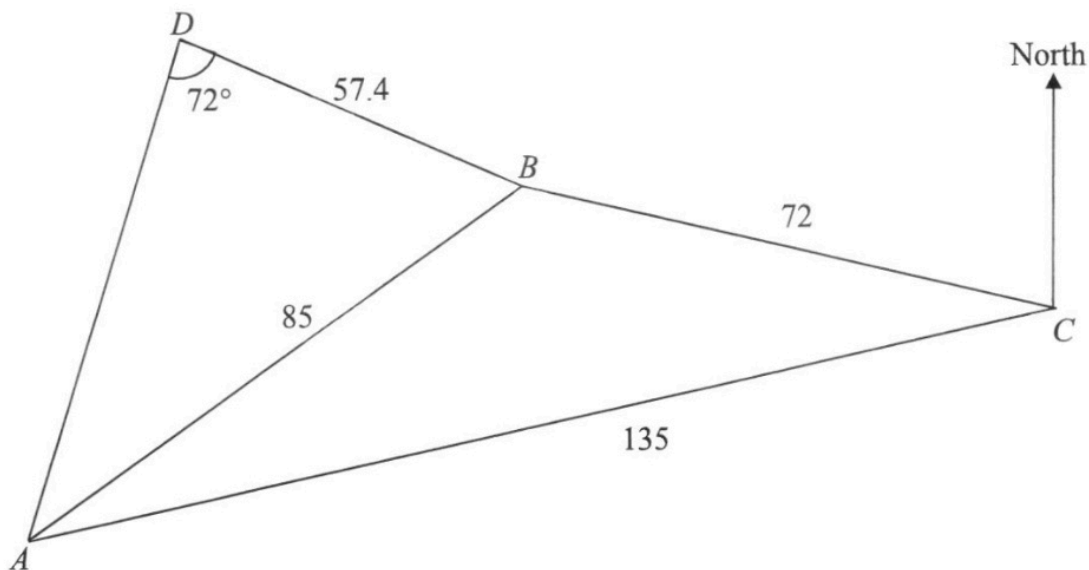




ABC and ADB are two triangular plots of land.
 $AB = 85$ m, $BC = 72$ m, $AC = 135$ and $BD = 57.4$ m.
 The bearing of B from C is 290° and angle $ADB = 72^\circ$.

- L1** (a) Find the bearing of A from B .

- L1** **8** **(b)** The plot of land ADB is to be filled with top soil. Calculate the area to be filled with top soil.

Answer m^2 [4]

- L2** **(c)** A 40 m antenna mast is installed vertically at point D . Find the largest angle of depression from the top of the mast to a point along the path AB .

Answer $^\circ$ [3] EM Notes

L1 (a) Find the bearing of A from B .

Answer 228.4° [4] EM Notes

- L1 8 (b)** The plot of land ADB is to be filled with top soil. Calculate the area to be filled with top soil.

$$\frac{\sin \angle DAB}{57.4} = \frac{\sin 72^\circ}{85}$$

$$\angle DAB = \sin^{-1} \left(\frac{57.4 \sin 72^\circ}{85} \right)$$

$$= 39.95925^\circ$$

$$\begin{aligned} \angle ABD &= 180 - 72 - 39.95925^\circ \text{ (sum of } \Delta) \\ &= 68.041^\circ \end{aligned}$$

$$\begin{aligned} \text{Area } \triangle ABD &= \frac{1}{2} \times 57.4 \times 85 \times \sin 68.041^\circ \\ &= 2262.514 \text{ m}^2 \\ &= 2260 \text{ m}^2 \end{aligned}$$

Answer 2260 m^2 [4]

- L2 (c)** A 40 m antenna mast is installed vertically at point D . Find the largest angle of depression from the top of the mast to a point along the path AB .

$$\sin 68.041^\circ = \frac{24}{57.4}$$

$$24 = 53.235629 \text{ m}$$

$$\tan \theta = \frac{40}{53.235629}$$

$$\theta = \tan^{-1} \left(\frac{40}{53.235629} \right)$$

$$= 36.9^\circ \text{ (l.d.p.)}$$

Answer 36.9 $^\circ$ [3] EM Notes

- 2 A box of chocolate contains 7 dark chocolates and 5 milk chocolates.
Sufyan takes a chocolate, selected at random, from the box and eats it.
LeLe then takes a chocolate, selected at random, from the box.

(i) Draw a tree diagram to show the probabilities of the possible outcomes.

Answer

[2]

(ii) Find, as a fraction in its simplest form, the probability that

(a) Sufyan and LeLe both picked dark chocolates,

Answer [1]

(b) LeLe picked a milk chocolate,

Answer [2]

(c) at least one milk chocolate was chosen.

Answer [2]

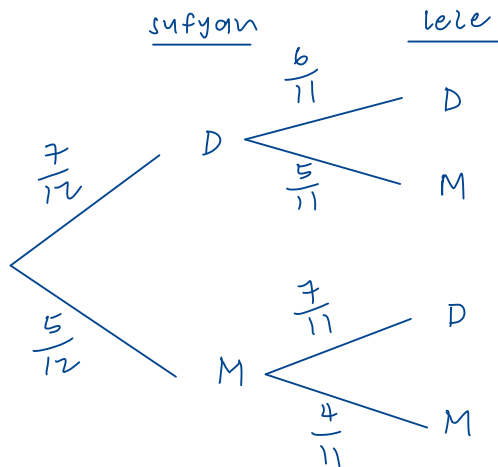
- 2 A box of chocolate contains 7 dark chocolates and 5 milk chocolates.
Sufyan takes a chocolate, selected at random, from the box and eats it.
LeLe then takes a chocolate, selected at random, from the box.

(i) Draw a tree diagram to show the probabilities of the possible outcomes.

Answer

let D and M
denote dark
and milk
chocolate
taken
respectively.

D - dark choco.
M - milk choco



[2]

(ii) Find, as a fraction in its simplest form, the probability that

(a) Sufyan and LeLe both picked dark chocolates,

$$P(\text{both D.C.}) = \frac{7}{12} \times \frac{6}{11}$$

$$= \frac{7}{22}$$

Answer $\frac{7}{22}$ [1]

(b) LeLe picked a milk chocolate,

$$P(\text{LeLe M}) = \left(\frac{7}{12} \times \frac{5}{11} \right) + \left(\frac{5}{12} \times \frac{4}{11} \right)$$

$$= \frac{5}{12}$$

Answer $\frac{5}{12}$ [2]

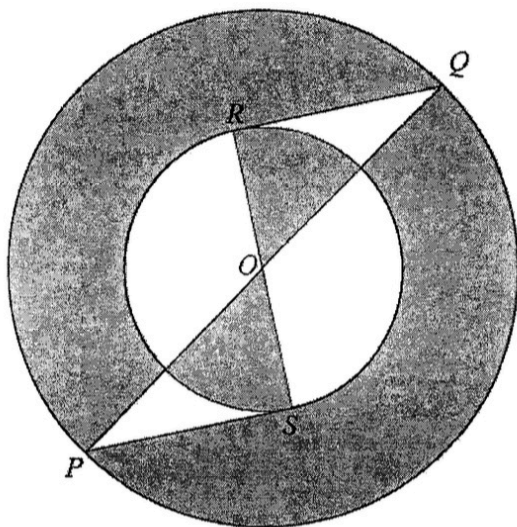
(c) at least one milk chocolate was chosen.

$$P(\text{at least 1 milk}) = 1 - P(\text{none milk})$$

$$= 1 - \left(\frac{7}{12} \times \frac{6}{11} \right)$$

$$= \frac{15}{22}$$

Answer $\frac{15}{22}$ [2]



The diagram shows two concentric circles with centre O .
 PQ is the diameter of the larger circle and RS is the diameter of the smaller circle.
 PS and RQ are tangents to the smaller circle.

- (a) Show that the triangle PSO is congruent to triangle QRO .
Give a reason for each statement you make.

Answer

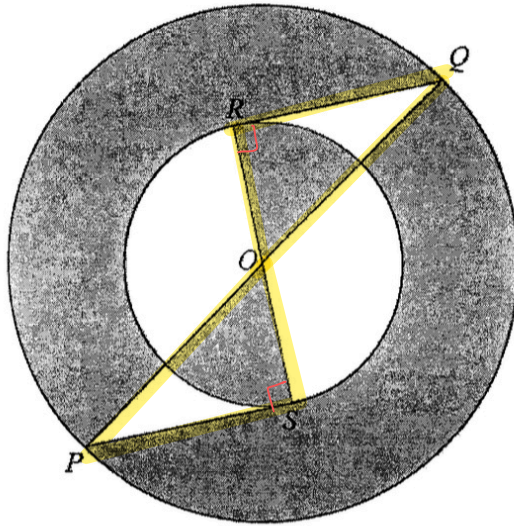
(b) The radius of the larger circle is 8 cm and the radius of the smaller circle is 4.36 cm.

(i) Calculate the area of triangle QRO .

Answer cm^2 [3]

(ii) Given that angle QOR is 0.995 radians, calculate the shaded area.

Answer cm^2 [4]



The diagram shows two concentric circles with centre O .
 PQ is the diameter of the larger circle and RS is the diameter of the smaller circle.
 PS and RQ are tangents to the smaller circle.

- (a) Show that the triangle PSO is congruent to triangle QRO .
 Give a reason for each statement you make.

Answer

In $\triangle PSO$ and $\triangle QRO$,
 $\angle PSO = \angle QRO = 90^\circ$ (tan $\perp$ rad)
 $PO = QO$ (radii of larger circle)
 $SO = RO$ (radii of smaller circle)

$\therefore \triangle PSO \equiv \triangle QRO$ (RHS Congruence Test)

or

In $\triangle PSO$ and $\triangle QRO$,
 $\angle POS = \angle QOR$ (vert. opp $\angle$)
 $OS = OR$ (radii of smaller circle)
 $OP = OQ$ (radii of larger circle)
 $\therefore \triangle PSO \equiv \triangle QRO$ (SAS Congruence Test)

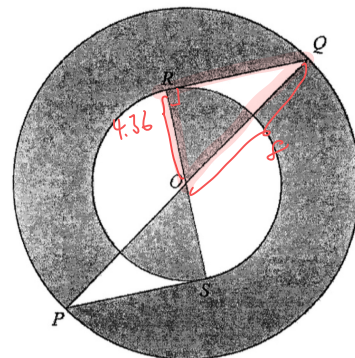
(b) The radius of the larger circle is 8 cm and the radius of the smaller circle is 4.36 cm.

(i) Calculate the area of triangle QRO .

$$QR^2 = 8^2 - 4.36^2 \text{ (Pythagoras' Theorem)}$$

$$QR = 6.7075$$

$$\begin{aligned} \text{Area } \triangle QRO &= \frac{1}{2} \times 6.7075 \times 4.36 \\ &= 14.672 \text{ cm}^2 \text{ (5.s.f)} \\ &= 14.6 \text{ cm}^2 \text{ (3.s.f)} \end{aligned}$$



Answer 14.6 cm² [3]

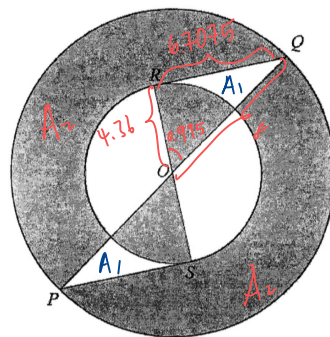
(ii) Given that angle QOR is 0.995 radians, calculate the shaded area.

$$\begin{aligned} \text{Area of 2 minor sector} &= 2 \left(\frac{1}{2} \times 4.36^2 \times 0.995 \right) \\ &= 2 (9.4573) \\ &= 18.915 \text{ cm}^2 \text{ (5.s.f)} \end{aligned}$$

$$\begin{aligned} \text{Area of unshaded } (A_1) &= (14.672 - 9.4573) \times 2 \\ &= 10.3294 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} A_2 &= \pi(8)^2 - \pi(4.36)^2 - 10.3294 \\ &= 131.01 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Total area} &= 131.01 + 18.915 \\ &= 150 \text{ cm}^2 \text{ (3.s.f)} \end{aligned}$$



Answer 150 cm² [4]

- 9 (a) A group of 50 patients had their blood pressure taken.

The results are shown in the table below.

Blood Pressure (mmHg)	$100 < x \leq 120$	$120 < x \leq 140$	$140 < x \leq 160$	$160 < x \leq 180$
Frequency	27	12	6	5

- (i) State the median class of blood pressure.

Answer [1]

- (ii) A blood pressure of 140 mmHg or higher indicates high blood pressure. Find the percentage of patients who may have high blood pressure.

Answer % [1]

- (iii) Calculate the standard deviation of the blood pressures.

Answer mmHg [2]

- (iv) The standard deviation of the blood pressures taken by a second group of patients was 22.1 mmHg.

Comment on one difference between the two distributions, making reference to this information.

Answer

 [1]

- 9 (a) A group of 50 patients had their blood pressure taken.

The results are shown in the table below.

Blood Pressure (mmHg)	$100 < x \leq 120$	$120 < x \leq 140$	$140 < x \leq 160$	$160 < x \leq 180$
Frequency	27	12	6	5

- (i) State the median class of blood pressure.

$$\text{median position} = \frac{50}{2} = 25^{\text{th}}$$

Answer $100 < x \leq 120$ [1]

- (ii) A blood pressure of 140 mmHg or higher indicates high blood pressure. Find the percentage of patients who may have high blood pressure.

$$\therefore = \frac{6+5}{50} \times 100$$

$$= 22\%$$

Answer 22 % [1]

- (iii) Calculate the standard deviation of the blood pressures.

if can use calculator to find answer

$$\bar{x} = \frac{27(110) + 12(130) + 6(150) + 5(170)}{50}$$

$$= 125.6$$

$$SD = \sqrt{\frac{27(110)^2 + 12(130)^2 + 6(150)^2 + 5(170)^2}{50} - (125.6)^2}$$

$$= 20.1 \text{ (3.s.f.)}$$

Answer 20.1 mmHg [2]

Need to show working
↓

- (iv) The standard deviation of the blood pressures taken by a second group of patients was 22.1 mmHg.

Comment on one difference between the two distributions, making reference to this information.

Answer The standard deviation of the first group of patient

is smaller than the second group. Hence, the blood pressure of the

first group of patients is more consistent than the second group. [1]

- (b) A drawer contains 2 blue socks and 6 white socks.
 Two socks are taken from the drawer at random without replacement.
 If the two socks are different colours, then a third sock is taken from the drawer.
 Otherwise, no third sock is taken.
- (i) Draw a tree diagram to show the probabilities of the possible outcomes.

Answer

[2]

- (ii) Find, as a fraction in its simplest form, the probability that

- (a) the first two socks taken are the same colour,

Answer [2]

- (b) a third sock is taken and it is the same colour as the first sock.

Answer [2]

- (b) A drawer contains 2 blue socks and 6 white socks.

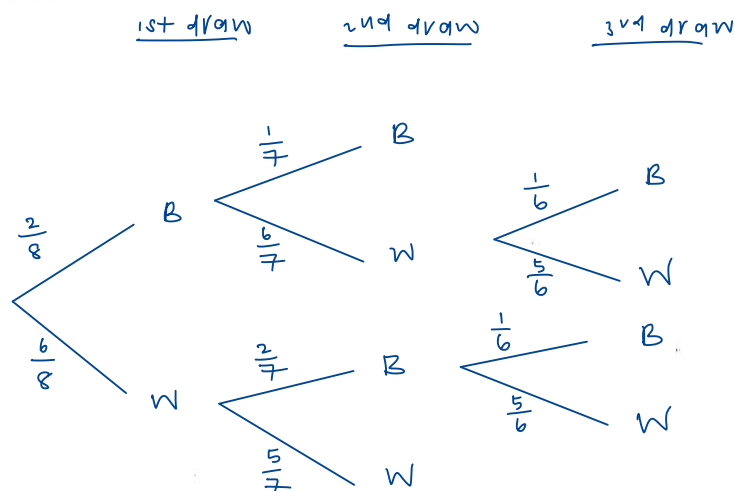
Two socks are taken from the drawer at random **without replacement**.

If the two socks are **different colours**, then **a third sock is taken** from the drawer.

Otherwise, no third sock is taken.

- (i) Draw a tree diagram to show the probabilities of the possible outcomes.

Answer



B - blue sock

W - white sock

[2]

- (ii) Find, as a fraction in its simplest form, the probability that

- (a) the first two socks taken are the same colour,

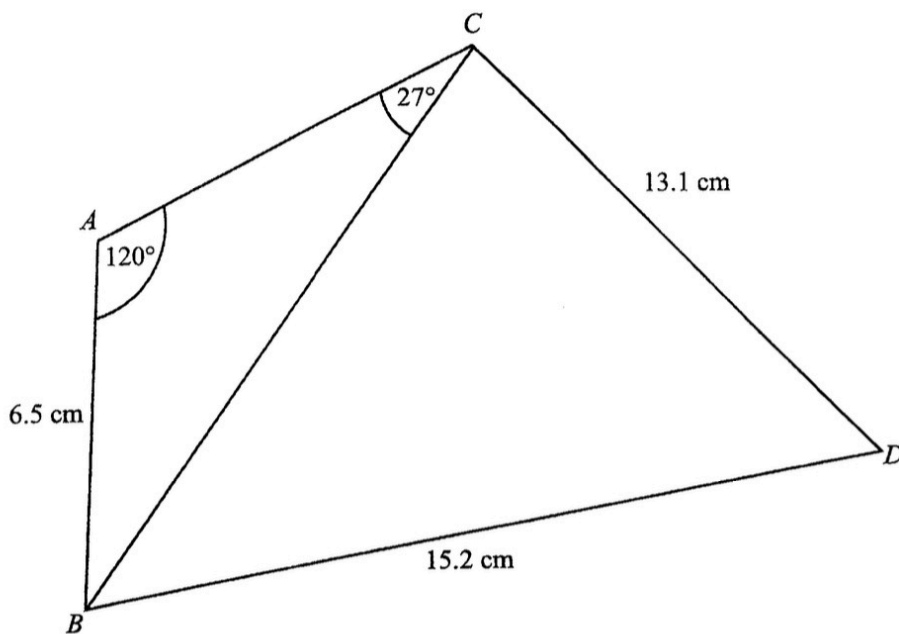
$$\begin{aligned}
 P(\text{same colour}) &= P(BB) + P(WW) \\
 &= \left(\frac{2}{8} \times \frac{1}{7}\right) + \left(\frac{6}{8} \times \frac{5}{7}\right) \\
 &= \frac{4}{7}
 \end{aligned}$$

Answer $\frac{4}{7}$ [2]

- (b) a third sock is taken and it is the same colour as the first sock.

$$\begin{aligned}
 &P(\text{3rd sock taken same colour as 1st sock}) \\
 &= P(BWB) + P(WBW) \\
 &= \left(\frac{2}{8} \times \frac{6}{7} \times \frac{5}{6}\right) + \left(\frac{6}{8} \times \frac{2}{7} \times \frac{5}{6}\right) \\
 &= \frac{3}{14}
 \end{aligned}$$

Answer $\frac{3}{14}$ [2]



In the quadrilateral, $AB = 6.5$ cm, $BD = 15.2$ cm and $CD = 13.1$ cm.
Angle $BAC = 120^\circ$ and angle $ACB = 27^\circ$.

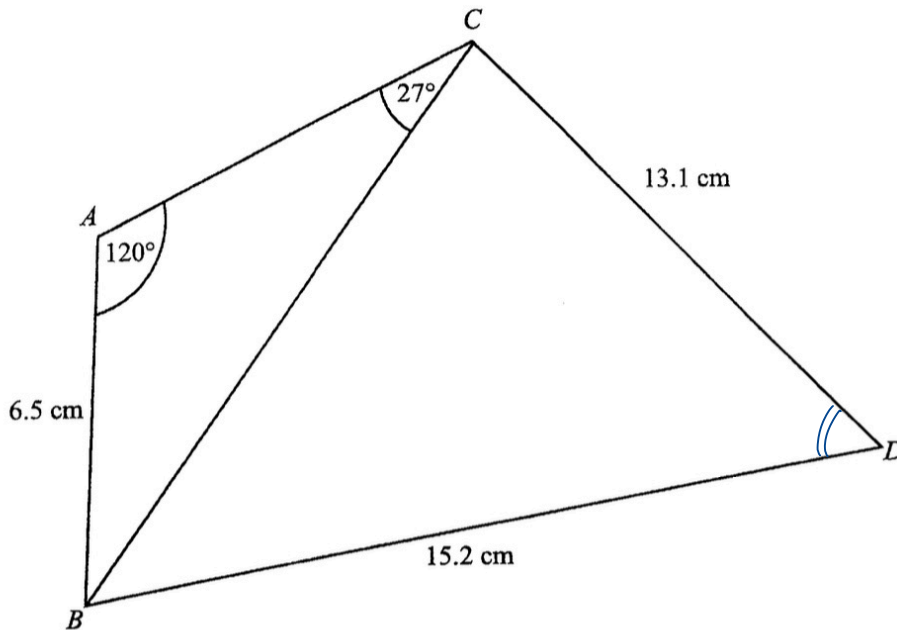
- (a) Find angle BDC .

Answer $^\circ$ [4]

- (b) Explain whether it is possible to draw a circle through the four vertices A , B , C and D .

.....

 [1]



In the quadrilateral, $AB = 6.5$ cm, $BD = 15.2$ cm and $CD = 13.1$ cm.
Angle $BAC = 120^\circ$ and angle $ACB = 27^\circ$.

(a) Find angle BDC .

$$\frac{BC}{\sin 120^\circ} = \frac{6.5}{\sin 27^\circ}$$

$$BC = \frac{6.5 \sin 120^\circ}{\sin 27^\circ}$$

$$= 12.399 \text{ cm}$$

$$12.399^2 = 15.2^2 + 13.1^2 - 2(15.2)(13.1) \cos \angle BDC$$

$$\angle BDC = \cos^{-1} \left(\frac{12.399^2 - 15.2^2 - 13.1^2}{-2(15.2)(13.1)} \right)$$

$$\angle BDC = 51.3^\circ \text{ (1 d.p.)} \quad \text{Answer } 51.3^\circ \quad [4]$$

(b) Explain whether it is possible to draw a circle through the four vertices A , B , C and D .

$$\angle BAD + \angle BDC = 120^\circ + 51.3^\circ = 171.3^\circ$$

$$\neq 180^\circ$$

$\therefore$ since opposite angles BAD and BDC are not supplementary, they are not angles in opposite segments. $\therefore$ It is not possible to draw a circle thru A, B, C, D . [1]

- (e) The cash price of a camera in Singapore is \$870.
- (i) The hire-purchase price of the camera is \$1056.
 The hire-purchase price is a deposit of 20% of the cash price plus
 12 equal monthly payments.
 Calculate one monthly payment.

Answer \$ [3]

- (ii) The price of the same camera in UK is £510.
 The exchange rate between Singapore dollars (\$) and UK pounds (£)
 is \$1 = £0.62.
 Calculate how much cheaper the camera is in the UK than in
 Singapore.

Answer \$ [2]

(e) The cash price of a camera in Singapore is \$870.

- (i) The hire-purchase price of the camera is \$1056.
The hire-purchase price is a deposit of 20% of the cash price plus 12 equal monthly payments.

Calculate **one** monthly payment.

$$\begin{aligned} \text{deposit} &= \frac{20}{100} \times \$870 \\ &= \$174 \end{aligned}$$

$$\begin{aligned} \text{total monthly payment} &= \$1056 - \$174 \\ &= \$882 \end{aligned}$$

$$\begin{aligned} \text{one month payment} &= \frac{\$882}{12} \\ &= \$73.50 \end{aligned}$$

Answer \$ 73.50 [3]

- (ii) The price of the same camera in UK is £510.
The exchange rate between Singapore dollars (\$) and UK pounds (£) is \$1 = £0.62.

Calculate how much cheaper the camera is in the UK than in Singapore.

$$\begin{aligned} &\times \frac{510}{0.62} \quad \left(\begin{array}{l} \$1 = £0.62 \\ \downarrow \times \frac{510}{0.62} \end{array} \right) \\ &\$822.58 = £510 \end{aligned}$$

$$\begin{aligned} \text{difference} &= \$870 - \$822.58 \\ &= \$47.42 \text{ (nearest cent)} \end{aligned}$$

Answer \$ 47.42 [2]

1 (a) Solve $\frac{1}{x-1} + \frac{2}{x-2} = \frac{3}{x-3}$.

Answer $x = \dots\dots\dots$ [2]

(b) Simplify $(a^2b)^{\frac{1}{2}} \times (ab^{-3})^{\frac{1}{3}}$.

Answer $\dots\dots\dots$ [1]

(c) (i) Express $x^2 + 11x - 15$ in the form $(x+a)^2 + b$.

Answer $\dots\dots\dots$ [1]

(ii) Hence solve the equation $x^2 + 11x - 15 = 0$, giving your answers correct to two decimal places.

Answer $x = \dots\dots\dots$ or $\dots\dots\dots$ [3]

(d) Simplify $\frac{3a^2 + 12a - 36}{ab - 2b + a^2 - 2a}$.

Answer [3]

1 (a) Solve $\frac{1}{x-1} + \frac{2}{x-2} = \frac{3}{x-3}$.

$$\frac{1(x-2) + 2(x-1)}{(x-1)(x-2)} = \frac{3}{x-3}$$

$$x = 1.5$$

$$\frac{x-2+2x-2}{(x-1)(x-2)} = \frac{3}{x-3}$$

$$\frac{3x-4}{x^2-3x+2} = \frac{3}{x-3}$$

$$(3x-4)(x-3) = 3x^2-9x+6$$

$$3x^2-13x+12 = 3x^2-9x+6$$

$$6 = 4x$$

Answer $x = \dots\dots\dots 1.5$ [2]

(b) Simplify $(a^2b)^{\frac{1}{2}} \times (ab^{-3})^{\frac{1}{3}}$.

$$= a^1 b^{\frac{1}{2}} \times \left(\frac{a}{b^3}\right)^{\frac{1}{3}}$$

$$= a b^{\frac{1}{2}} \times \frac{a^{\frac{1}{3}}}{b}$$

$$= \frac{a^{\frac{4}{3}}}{b^{\frac{1}{2}}}$$

Answer $\dots\dots\dots \frac{a^{\frac{4}{3}}}{b^{\frac{1}{2}}}$ [1]

(c) (i) Express $x^2+11x-15$ in the form $(x+a)^2+b$.

$$= \left(x + \frac{11}{2}\right)^2 - \left(\frac{11}{2}\right)^2 - 15$$

$$= \left(x + \frac{11}{2}\right)^2 - 45\frac{1}{4}$$

Answer $\left(x + \frac{11}{2}\right)^2 + 45\frac{1}{4}$ [1]

(ii) Hence solve the equation $x^2+11x-15=0$, giving your answers correct to two decimal places.

$$\left(x + \frac{11}{2}\right)^2 - 45\frac{1}{4} = 0$$

$$\left(x + \frac{11}{2}\right)^2 = 45\frac{1}{4}$$

$$x + \frac{11}{2} = \pm \sqrt{45\frac{1}{4}}$$

$$x = \sqrt{45\frac{1}{4}} - \frac{11}{2} \quad \text{or} \quad x = -\sqrt{45\frac{1}{4}} - \frac{11}{2}$$

$$= 1.23 \text{ (2.d.p.)} \quad x = -12.23 \text{ (2.d.p.)}$$

Answer $x = \dots\dots\dots 1.23$ or $\dots\dots\dots -12.23$ [3]

(d) Simplify $\frac{3a^2+12a-36}{ab-2b+a^2-2a}$

$$= \frac{3(a^2+4a-12)}{b(a-2)+a(a-2)}$$

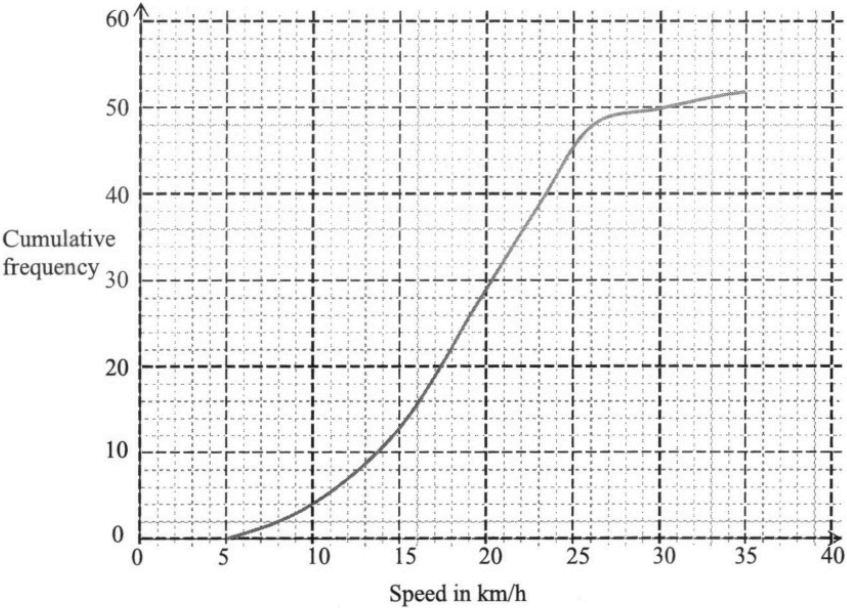
$$= \frac{3(\cancel{a-2})(a+6)}{\cancel{a-2}(a+b)}$$

$$= \frac{3(a+6)}{a+b}$$

$$= \frac{3a+18}{a+b}$$

Answer $\frac{3a+18}{a+b}$ [3]

- 3 The cumulative frequency graph below shows speeds in km/h of cyclists passing a certain point during a race.



(a) Use the curve to estimate

- (i) the median speed of the cyclists,

Answer km/h [1]

- (ii) the percentage of cyclists having speed more than 21 km/h.

Answer % [1]

- (iii) the ninety-sixth percentile.

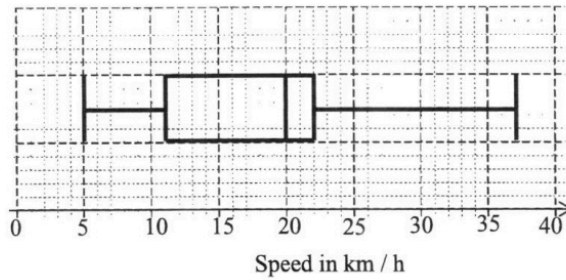
Answer km/h [1]

- (iv) Complete the grouped frequency table of the speeds of the cyclists passing a certain point during the race.

Speed (x km/h)	$5 \leq x < 10$	$10 \leq x < 15$	$15 \leq x < 20$	$20 \leq x < 25$	$25 \leq x < 30$	$30 \leq x < 35$
Frequency	4					

[2]

- (b) In the previous year, the same race had taken place.
The box-and-whisker plot shows the distribution of the speed of the cyclists.



Make two comparisons between the performances of the cyclists in the two years.
Use figures to support your answer.

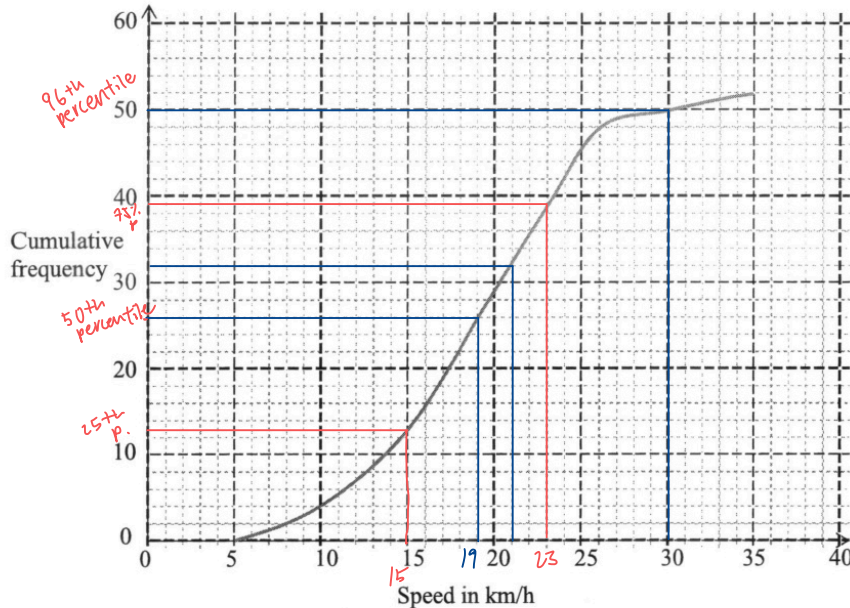
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.....

.....

..... [3]

- 3 The cumulative frequency graph below shows speeds in km/h of cyclists passing a certain point during a race.



Total = 52 cyclists

(a) Use the curve to estimate

- (i) the median speed of the cyclists,

$$50^{\text{th}} = 52 \times \frac{1}{2} = 26$$

Answer 19 km/h [1]

- (ii) the percentage of cyclists having speed more than 21 km/h.

$$52 - 32 = 20$$

$$\therefore \frac{20}{52} \times 100\% = 38.5\% \text{ (3.s.f.)}$$

Answer 38.5 % [1]

- (iii) the ninety-sixth percentile.

$$96^{\text{th}} \text{ percentile} = 52 \times 96\% = 49.92 \approx 50$$

Answer 30 km/h [1]

- (iv) Complete the grouped frequency table of the speeds of the cyclists passing a certain point during the race.

Speed (x km/h)	$5 \leq x < 10$	$10 \leq x < 15$	$15 \leq x < 20$	$20 \leq x < 25$	$25 \leq x < 30$	$30 \leq x < 35$
Frequency	4	9	16	16/17	5/4	2

[2]

Total must add up to 52

- (b) In the previous year, the same race had taken place.
The box-and-whisker plot shows the distribution of the speed of the cyclists.

previous yr

$$\text{median} = 20 \text{ km/h}$$

$$\text{IQR} = 22 - 11$$

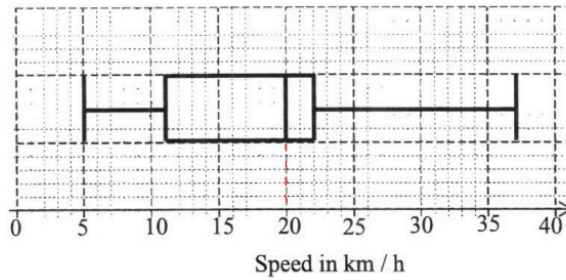
$$= 11 \text{ km/h}$$

current yr

$$\text{median} = 19 \text{ km/h}$$

$$\text{IQR} = 23 - 15$$

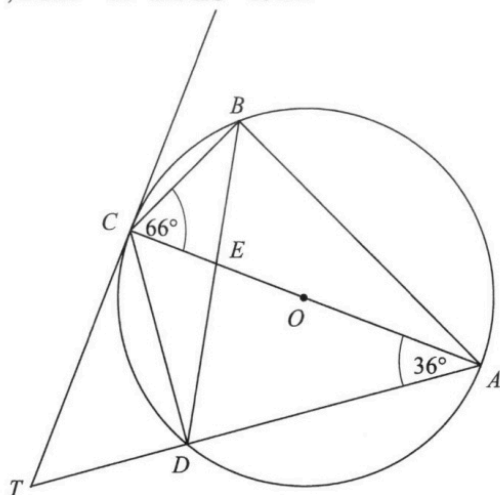
$$= 8 \text{ km/h}$$



Make two comparisons between the performances of the cyclists in the two years.
Use figures to support your answer.

Median of previous year is higher than current year of cyclist. Hence, on average the previous year of cyclists performed better.
IQR of current year is lower than previous year of cyclist.
Hence performances of current year's cyclist is more consistent. [3]

- 6 In the diagram, A, B, C and D are points on the circle with centre O .
 The lines AC and BD intersect at E .
 The tangent to the circle at C meets AD produced at T .
 $\angle ACB = 66^\circ$, $\angle CAD = 36^\circ$ and $AC = 10$ cm.



- (a) Find the angle DCT .

Answer [2]

- (b) Show that the triangles BCE and ADE are similar.
 Give a reason for each statement you make.

.....

.....

.....

.....

..... [2]

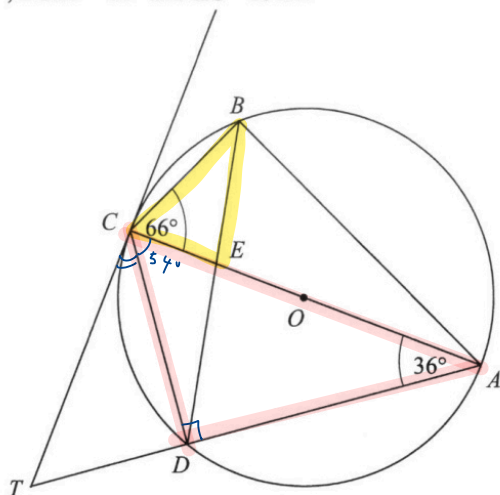
- (c) Calculate DE .

Answer cm [4]

- (d) Calculate the length of the minor arc BC .

Answer cm [3]

- 6 In the diagram, A, B, C and D are points on the circle with centre O . The lines AC and BD intersect at E . The tangent to the circle at C meets AD produced at T . $\angle ACB = 66^\circ$, $\angle CAD = 36^\circ$ and $AC = 10$ cm.



- (a) Find the angle DCT .

$$\begin{aligned}\angle CDA &= 90^\circ \text{ (right } \angle \text{ in a semicircle)} \\ \angle DCA &= 180^\circ - 90^\circ - 36^\circ \text{ (}\angle \text{ sum of } \triangle\text{)} \\ &= 54^\circ \\ \angle ACT &= 90^\circ \text{ (tgn } \perp \text{ rad)} \\ \angle DCT &= 90^\circ - 54^\circ \\ &= 36^\circ\end{aligned}$$

Answer 36° [2]

- (b) Show that the triangles BCE and ADE are similar. Give a reason for each statement you make.

$$\begin{aligned}\angle BEC &= \angle AED \text{ (vert. opp. } \angle\text{)} \\ \angle CBE &= \angle DAE \text{ (}\angle \text{ in the same segment)} \\ &= 36^\circ \\ \therefore \triangle BCE \text{ and } \triangle ADE \text{ are similar (AA similarity test) [shown]}\end{aligned}$$

[2]

(c) Calculate DE .

$$\cos 36^\circ = \frac{DE}{10}$$

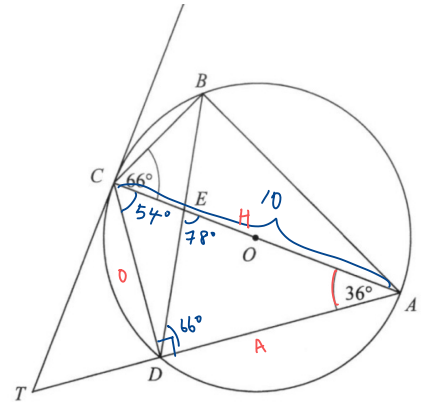
$$DE = 10 \cos 36^\circ$$

$$\angle DEA = 180^\circ - 36^\circ - 66^\circ \text{ (sum of } \Delta) \\ = 78^\circ$$

$$\frac{DE}{\sin 36^\circ} = \frac{DA}{\sin 66^\circ}$$

$$DE = \frac{DA \sin 36^\circ}{\sin 66^\circ}$$

$$DE = 4.8615 \\ = 4.86 \text{ cm}$$



Answer 4.86 cm [4]

(d) Calculate the length of the minor arc BC .

★★ To calculate arc length in degree mode $= \frac{\theta}{360^\circ} \times 2\pi r$ pls memorise $\pi \times \pi$

$$\angle CBA = 90^\circ \text{ (right } \angle \text{ in a semicircle)}$$

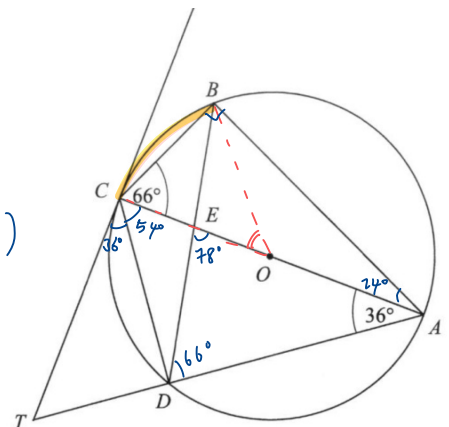
$$\angle BAC = 180^\circ - 90^\circ - 66^\circ \text{ (sum of } \Delta)$$

$$= 24^\circ$$

$$\angle BOC = 24^\circ \times 2 \text{ (} \angle \text{ at centre} = 2 \angle \text{ at circumference)}$$

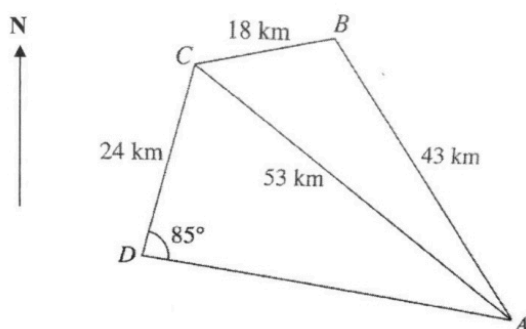
$$= 48^\circ$$

$$\text{length of minor arc } BC = \frac{48^\circ}{360^\circ} \times 2\pi \times 5 \\ = 4.19 \text{ cm (3.s.f.)}$$



Answer 4.19 cm [3]

- 7 A city tour starting at Hotel A is planned. The coach travels 43 km in the **north-west** direction to B , the zoo, where it stops for 3 hours. After the visit to the zoo, the coach continues its journey to a popular seafood restaurant at C , 18 km away, for lunch. After lunch, the coach brings the tourists to visit a museum at D , 24 km away and $\angle CDA = 85^\circ$.



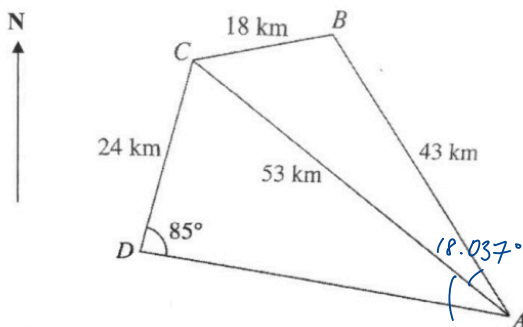
- (a) Find
(i) $\angle BAC$,

Answer [2]

- (ii) $\angle CAD$.

Answer [2]

- 7 A city tour starting at Hotel A is planned. The coach travels 43 km in the north-west direction to B , the zoo, where it stops for 3 hours. After the visit to the zoo, the coach continues its journey to a popular seafood restaurant at C , 18 km away, for lunch. After lunch, the coach brings the tourists to visit a museum at D , 24 km away and $\angle CDA = 85^\circ$.



- (a) Find
(i) $\angle BAC$,

$$18^2 = 53^2 + 43^2 - 2(53)(43) \cos \angle BAC$$

$$\angle BAC = \cos^{-1} \left(\frac{18^2 - 53^2 - 43^2}{-2(53)(43)} \right)$$

$$= 18.037^\circ$$

$$= 18.0^\circ (1.d.p)$$

Answer 18.0° [2]

- (ii) $\angle CAD$.

$$\frac{\sin \angle CAD}{24} = \frac{\sin 85^\circ}{53}$$

$$\sin \angle CAD = \frac{24 \sin 85^\circ}{53}$$

$$\angle CAD = \sin^{-1} \left(\frac{24 \sin 85^\circ}{53} \right)$$

$$= 26.814^\circ$$

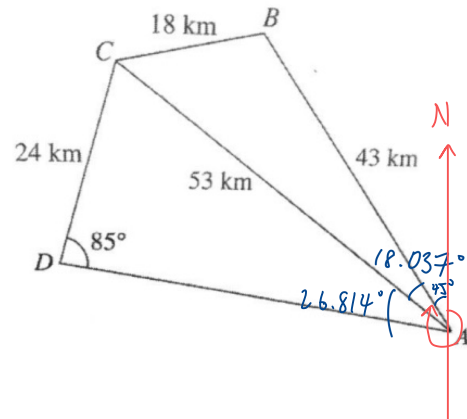
$$= 26.8^\circ (1.d.p)$$

Answer 26.8° [2]

- (b) Find the bearing of C from A.

$$\angle NAB = 45^\circ$$

$$\begin{aligned}\angle NAC &= 360^\circ - 45^\circ - 18.037^\circ \text{ (Lat a pt)} \\ &= 297.0^\circ \text{ (1.d.p.)}\end{aligned}$$



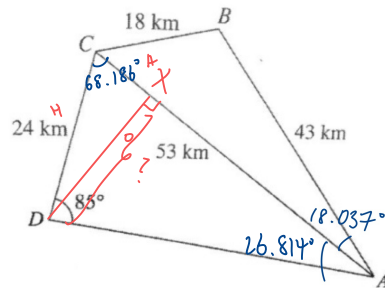
Answer 297.0° [2]

- (c) Calculate the shortest distance from D to AC.

$$\begin{aligned}\angle DAC &= 180^\circ - 85^\circ - 26.814^\circ \text{ (Sum of } \Delta) \\ &= 68.186^\circ\end{aligned}$$

$$\sin 68.186^\circ = \frac{DX}{24}$$

$$\begin{aligned}DX &= 24 \sin 68.186^\circ \\ &= 22.3 \text{ km (3.s.f.)}\end{aligned}$$



Answer 22.3 km [3]

- 8 (a) $\xi = \{\text{integers } x: 0 < x < 12\}$
 $P = \{\text{even numbers}\}$
 $Q = \{\text{prime numbers}\}$

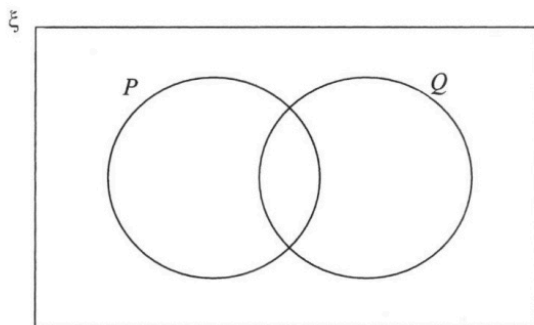
(i) List the elements in P .

Answer [1]

(ii) List the elements in $(P \cup Q)'$.

Answer [1]

(iii) Complete the Venn diagram showing all the elements of ξ .



[1]

(iv) Find the value of

(a) $n[(P \cap Q) \cup Q']$

Answer [1]

(b) $n[(P \cup Q) \cap (P \cap Q)']$

Answer [1]

- 8 (a) $\xi = \{\text{integers } x: 0 < x < 12\}$
 $P = \{\text{even numbers}\} \quad \{2, 4, 6, 8, 10\}$
 $Q = \{\text{prime numbers}\} \quad \{2, 3, 5, 7, 11\}$

(i) List the elements in P .

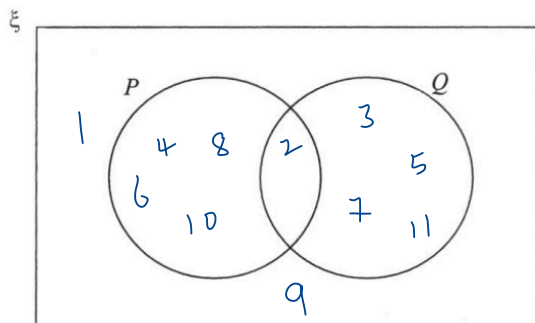
Answer 2, 4, 6, 8, 10 [1]

(ii) List the elements in $(P \cup Q)'$.



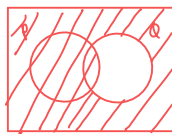
Answer 1, 9 [1]

(iii) Complete the Venn diagram showing all the elements of ξ .



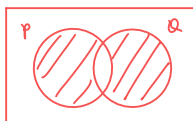
[1]

- (iv) Find the value of
 (a) $n[(P \cap Q) \cup Q']$



Answer 7 [1]

- (b) $n[(P \cup Q) \cap (P \cap Q)']$



Answer 8 [1]

- 8 (a) $\xi = \{\text{integers } x: 0 < x < 12\}$
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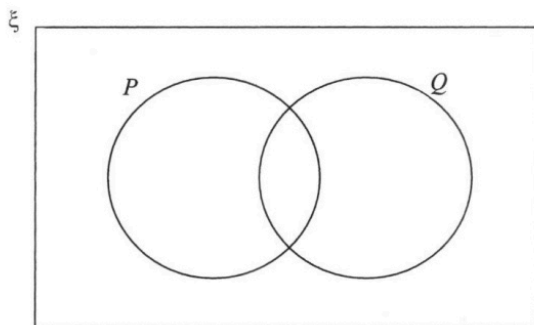
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